

Foundation of Programming

Assignment No. 16

PROBLEM STATEMENT:

To accept a list of N integers and partition the list into two sublists containing even and odd numbers.

OBJECTIVE:

1. To understand how to work with arrays in C.
2. To learn how to separate numbers based on conditions.
3. To practice the use of loops and conditional statements.
4. To display two different lists for even and odd numbers.

THEORY:

In C programming, arrays are used to store multiple values of the same data type. By using loops and conditional statements, it is possible to examine each element in the array and classify it based on certain conditions. In this program, the user enters N integers which are stored in an array. Each number is then checked using the modulus operator (%). If a number is divisible by 2, it is categorized as an even number; otherwise, it is an odd number. The program then displays two separate lists: one for even numbers and another for odd numbers.

ALGORITHM:

- STEP 1: START
- STEP 2: Declare a variable N
- STEP 3: Accept the value of N from the user
- STEP 4: Declare an array arr[] of size N
- STEP 5: Accept the elements of arr[]
- STEP 6: For every element of arr[] check if %2 == 0
 If yes, add to even[]
 Else, add to odd[]
- STEP 7: Display the elements of even[] and odd[]
- STEP 8: STOP

PLATFORM : Linux

CODE:

```
#include <stdio.h>

int main() {
    int n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n], even[n], odd[n];
    int e = 0, o = 0;

    printf("Enter %d integers:\n", n);
    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);

        if(arr[i] % 2 == 0) {
            even[e++] = arr[i];
        } else {
            odd[o++] = arr[i];
        }
    }

    printf("\nEven numbers:\n");
    for(i = 0; i < e; i++) {
        printf("%d ", even[i]);
    }

    printf("\nOdd numbers:\n");
    for(i = 0; i < o; i++) {
        printf("%d ", odd[i]);
    }

    return 0;
}
```

OUTPUT:

```
Enter number of elements: 10
Enter 10 integers:
1 5 3 7 2 8 8 9 4 0

Even numbers:
2 8 8 4 0
Odd numbers:
1 5 3 7 9
```

CONCLUSION:

Thus, the C program successfully accepts a list of integers and partitions them into two separate lists of even and odd numbers.